Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit*

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July 2026

Abstract

A two-parameter family of transition kernels satisfies the Chapman–Kolmogorov property exactly when it defines a flat transport over the complex of time intervals. This paper develops the geometry of families for which flatness fails. The defect $\Delta(t, s, r)$ is not an artifact of a local approximation scheme but the intrinsic failure of the Chapman–Kolmogorov property; it is a kernel-valued 2-cochain, and associativity of composition imposes on it a coherence identity of Bianchi type. A first result is negative and closes the natural route: under re-dressings acting by conjugation subject to positivity and mass conservation, the orbit closure of every family contains flat families, so no gauge-orbit minimization defines an invariant. The invariant is instead built from dilations. Within a declared dilation category, let $\mathcal{F}_d$ denote the compact set of compressions of flat transports on d states. The profile $h_d(V)$, the attained distance from V to $\mathcal{F}_d$ in a metric on families weighted by the law of the process, is nonincreasing in d, and its zero stratum consists exactly of the families that extend to a flat transport on a d-state structured enlargement. On a finite time grid the profile terminates at zero precisely for the conditional families of genuine processes, and the minimal terminating dimension defines the memory rank. First-order theory at the projection is developed on the tangent and normal cones of the strata. The profile is category-relative: enlarging the category, for instance from time-homogeneous to time-inhomogeneous generators, strictly deforms it. Explicit computations for finite-state families recover the hidden dimension of lumped Markov chains exactly.

Keywords. Chapman–Kolmogorov divisibility, indivisible stochastic processes, hidden Markov models, semialgebraic sets, metric algebraic geometry, realization theory, memory rank

AMS subject classifications. 60J10, 14P10, 93B15, 90C26, 15B51

Significance statement

Coarse observation of a Markov system generically produces a transition family that no longer composes: the two-time kernels violate the Chapman–Kolmogorov relation, and no generator reproduces them. This paper measures that failure geometrically. It first proves that the natural gauge-theoretic route to an invariant is empty, because every family can be re-dressed arbitrarily close to a memoryless one, and then constructs the correct invariant as a metric projection: the distance from the observed family to the compact semialgebraic set of families explainable by a d-state hidden Markov model. The resulting profile in d is monotone, Lipschitz-stable in the data,

*This work was supported by Adagio Institute, Inc.

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and vanishes exactly at the minimal hidden dimension, which the paper names the memory rank. The construction places the analysis of non-Markovian transition data inside metric algebraic geometry, where distances to model varieties, their critical points, and their stratifications are native objects, and it supplies realization theory with a quantitative, metric refinement of the classical rank-based dimension counts.

1 Introduction

Let $\{V(t \leftarrow s)\}_{s \le t}$ be a two-time family of column-stochastic transition kernels on a finite state space, indexed by a finite grid of times. If the family arises as the conditional transition probabilities of a Markov chain observed in full, it composes:

$$V(t \leftarrow s) \;=\; V(t \leftarrow r)\, V(r \leftarrow s), \qquad s \le r \le t, \tag{1}$$

which is the Chapman–Kolmogorov property, and which we read geometrically as the statement that V defines a flat transport over the complex of time intervals. If instead the family arises by observing a Markov system through a lumping of its states, the property (1) generically fails [10]. The failure is quantified by the defect

$$\Delta_V(t, s, r) \;=\; V(t \leftarrow s) - V(t \leftarrow r)\, V(r \leftarrow s), \qquad s < r < t, \tag{2}$$

the object called the indivisibility deficit in the program of which this paper is a part [17, 18, 19, 20], and, in the quantum-information literature on non-Markovian dynamics, the object whose size measures the failure of divisibility of a dynamical map [14, 4, 13]. Proposition 3.2 of [18] shows that a family violating (1) is the two-time evolution of no Markov semigroup on the same state space; the deficit is therefore not an artifact of a local approximation scheme, in contrast with the germ defects of sewing theory [8, 7, 11, 5, 12], but an intrinsic property of the observed family. The question the paper answers is how to convert Δ_V into a well-defined, stable, computable invariant.

Two features of the answer are dictated by failed alternatives, and the paper records the failures as theorems because they close routes a reader will otherwise propose. First, the deficit cannot be normalized away: re-dressings of the family by invertible changes of description act on Δ_V by conjugation and never annihilate it (Lemma 3.2). Second, and less obviously, the deficit cannot be minimized over such re-dressings either: Theorem 3.3 shows that the closure of the gauge orbit of *every* family contains flat families, so the orbital infimum of any continuous flatness-distance is identically zero and carries no information. The mechanism is a one-parameter collapse onto rank-one reset kernels, and Remark 3.5 places it where it belongs, in the instability theory of a non-reductive group action. The gauge route to an invariant is empty.

The invariant is instead a metric projection. Fix a dilation category: a class of hidden Markov lifts, each consisting of a flat transport on d hidden states together with a labeled surjection onto the observed states, and let $\mathcal{F}_d$ be the closure of the set of observable compressions of such lifts (Definition 4.2). The set $\mathcal{F}_d$ is compact and semialgebraic (Propositions 4.3 and 4.5), the sets are nested in d (Proposition 4.4), and the liftability profile

$$h_d(V) \;=\; \operatorname{dist}\big(V, \mathcal{F}_d\big) \tag{3}$$

is attained, nonincreasing in d, and 1-Lipschitz in the family (Proposition 4.6), which is the property that makes its plug-in estimation from data principled. Its zero stratum is exactly the set of families that extend to a flat transport on a d-state structured enlargement, its minimal vanishing dimension defines the memory rank $\mathrm{rk}_{\mathrm{mem}}(V)$ (Definition 5.1), and Theorem 5.7 shows that on a finite grid

the profile terminates at zero precisely for the conditional families of genuine processes, with an explicit dimension bound; families that are conditional families of no process are separated from every $\mathcal{F}_d$ by a computable marginal-consistency defect (Theorem 5.6). First-order theory at the projection is developed on the tangent and normal cones of the strata (Section 6), and the profile is category-relative in a quantifiable way: enlarging the category from time-homogeneous to time-inhomogeneous lifts weakly decreases it and can strictly decrease it (Proposition 7.1). Section 8 reports finite-state computations in which the profile recovers the hidden dimension of lumped Markov chains exactly and exhibits the strictly positive plateau of non-realizable families.

The paper's tools and its intended audience are those of metric algebraic geometry [3]: the model sets $\mathcal{F}_d$ are compact semialgebraic images of parameter spaces, the invariant is a Euclidean distance to a model set, its critical-point structure is governed by the Euclidean distance degree [6], and its regularity theory lives on Whitney stratifications of semialgebraic sets [2] through the normal cones of variational analysis [15]. The memory rank refines a classical circle of ideas. In the linear category, the minimal dimension of a realization of an input-output map is a Hankel rank, computed by the Ho–Kalman construction [9]; for hidden Markov models the corresponding realization problem is long-studied and known to be delicate precisely because of positivity [1, 21]; and the general phenomenon that the minimal size of a lift depends on the positivity structure of the lifting category is the content of extension-complexity theory, where lifts of the same object through different cones have provably different minimal dimensions [22]. The memory rank is the metric, temporal member of this family: it measures not whether a d-dimensional explanation exists but how far the data sit from the nearest one, and Proposition 5.3 calibrates it against the deficit norms of [18, 20] through a two-sided Łojasiewicz comparison.

The contributions, in the order of the paper: (i) the coherence identity of Bianchi type that associativity forces on the deficit, with a two-line proof (Proposition 2.2); (ii) the gauge trivialization theorem and its invariant-theoretic reading (Theorem 3.3, Remark 3.5); (iii) the construction of the liftability profile with compactness, attainment, monotonicity, semialgebraicity, stability, and weight-independence of the strata (Section 4); (iv) memory rank, the realization theorem with dimension bound, and the realizability gap (Section 5); (v) first-order theory at the projection on stratified model sets (Section 6); (vi) category dependence with a quantitative example (Section 7); and (vii) exact finite-state computations (Section 8). Section 9 states open problems, including the conjectured bridge to the selection functionals of [20].

2 Transition families and the deficit complex

2.1 Families, metric, and flatness

Fix an integer $n \geq 2$, a finite time grid $\mathbb{T} = \{t_0 < t_1 < \cdots < t_m\}$ with $m \geq 2$, and write i for t_i when no confusion arises. Let $\mathcal{S}_n$ denote the set of column-stochastic $n \times n$ matrices, $A \geq 0$ entrywise and $\mathbf{1}^\top A = \mathbf{1}^\top$. A *transition family* is an element

$$V = \{V(j \leftarrow i)\}_{0 \leq i < j \leq m} \ \in \ \mathcal{V} := \prod_{i<j} \mathcal{S}_n.$$

The convention matches [18, 20], where the family is written $\Gamma(t \leftarrow s)$; we write V throughout. The space $\mathcal{V}$ is a compact convex subset of a finite-dimensional real vector space.

Fix once and for all strictly positive weights: numbers $p_{ij} > 0$ for $i < j$ and vectors $\pi_i \in \mathbb{R}^n_{>0}$. These are supplied in applications by the law of the observed process, for instance π_i the occupation

law at time t_i, and are held fixed thereafter. They induce the inner product and norm

$$\langle A, B\rangle \;=\; \sum_{i<j} p_{ij} \sum_{a,b=1}^{n} \pi_i(a)\, A(j \leftarrow i)_{ba}\, B(j \leftarrow i)_{ba}, \qquad \|A\| = \langle A, A\rangle^{1/2}, \tag{4}$$

on the ambient space of kernel families, and the metric $\mathrm{dist}(V, W) = \|V - W\|$ on $\mathcal{V}$. All norms on the ambient space are equivalent; Lemma 4.8 records that the structural objects of the paper do not depend on the choice of weights.

Definition 2.1. $V \in \mathcal{V}$ is *flat* if (1) holds for all $i \le k \le j$ in the grid. Equivalently, $V(j \leftarrow i) = M_j M_{j-1} \cdots M_{i+1}$ where $M_k := V(k \leftarrow k{-}1) \in \mathcal{S}_n$. The set of flat families is denoted $\mathcal{F}\mathrm{lat} \subset \mathcal{V}$; it is the compact zero set of the polynomial system $\Delta_V = 0$ and is therefore closed.

2.2 The deficit as a 2-cochain and its coherence identity

For $s < r < t$ in the grid define $\Delta_V(t, s, r)$ by (2). Since $\mathbf{1}^\top V = \mathbf{1}^\top$ for every kernel, $\mathbf{1}^\top \Delta_V(t, s, r) = 0$: the deficit takes values in the space of matrices with vanishing column sums. We regard the assignment $(t, s, r) \mapsto \Delta_V(t, s, r)$ as a kernel-valued 2-cochain on the order complex of the grid, the family V itself being the 1-cochain on intervals whose failure to be multiplicative the deficit measures. Associativity of matrix multiplication forces a syzygy.

Proposition 2.2 (Coherence identity)**.** *For all $s < r < q < t$ in the grid,*

$$\Delta_V(t, s, r) \;-\; \Delta_V(t, s, q) \;-\; V(t \leftarrow q)\, \Delta_V(q, s, r) \;+\; \Delta_V(t, r, q)\, V(r \leftarrow s) \;=\; 0. \tag{5}$$

Proof. By (2), $\Delta_V(t,s,r) - \Delta_V(t,s,q) = V(t \leftarrow q)V(q \leftarrow s) - V(t \leftarrow r)V(r \leftarrow s)$. Substitute $V(q \leftarrow s) = \Delta_V(q,s,r) + V(q \leftarrow r)V(r \leftarrow s)$ and $V(t \leftarrow r) = \Delta_V(t,r,q) + V(t \leftarrow q)V(q \leftarrow r)$; the two triple products $V(t \leftarrow q)V(q \leftarrow r)V(r \leftarrow s)$ cancel, leaving $V(t \leftarrow q)\Delta_V(q,s,r) - \Delta_V(t,r,q)V(r \leftarrow s)$. □

Remark 2.3. Identity (5) is the statement $\delta_V \Delta_V = 0$ for the V-twisted coboundary $(\delta_V \Phi)(t,s,r,q) = V(t \leftarrow q)\Phi(q,s,r) - \Phi(t,s,q) + \Phi(t,s,r) - \Phi(t,r,q)V(r \leftarrow s)$ on kernel-valued cochains, the analog of the Bianchi identity satisfied by a curvature. It is also the four-point instance of the coherence tower governing homotopy-associative structures, in which the deficit plays the associator and (5) the pentagon of the associahedron K_4 [16]; the higher identities on five and more points follow from (5) by the same substitution scheme and are not used in this paper. In the computations of Section 8 the identity holds to machine precision on every family tested and serves as a correctness check on any implementation that estimates deficits from data.

3 The gauge route and its trivialization

The natural first attempt at an invariant declares part of the deficit to be an artifact of description and removes it by re-dressing. This section defines the re-dressings, proves that they act on the deficit by conjugation and can never annihilate it, and then proves the sharper negative result: they can shrink it, in norm, to zero, for every family, so that the orbital infimum of any continuous flatness-distance is identically zero. The construction is completely explicit.

Definition 3.1. A *re-dressing* is a tuple $g = (g_0, \ldots, g_m)$ of invertible real $n \times n$ matrices with $\mathbf{1}^\top g_i = \mathbf{1}^\top$ for every i. It acts on a family by

$$V^g(j \leftarrow i) \;=\; g_j\, V(j \leftarrow i)\, g_i^{-1}.$$

Since $\mathbf{1}^\top g_j = \mathbf{1}^\top$ implies $\mathbf{1}^\top g_j^{-1} = \mathbf{1}^\top$, column sums are preserved: $\mathbf{1}^\top V^g = \mathbf{1}^\top$. The re-dressing is *admissible for* V if in addition $V^g(j \leftarrow i) \geq 0$ entrywise for all $i < j$, so that $V^g \in \mathcal{V}$. The *orbit* of V is $\mathcal{O}(V) = \{V^g : g \text{ admissible for } V\} \subseteq \mathcal{V}$.

Lemma 3.2 (Covariance)**.** *For every re-dressing g admissible for V and all $s < r < t$,*

$$\Delta_{V^g}(t, s, r) \;=\; g_t \, \Delta_V(t, s, r) \, g_s^{-1}.$$

In particular $\Delta_{V^g} = 0$ if and only if $\Delta_V = 0$: no re-dressing annihilates a nonzero deficit, and re-dressings map $\mathcal{F}\mathrm{lat}$ to $\mathcal{F}\mathrm{lat}$.

Proof. $V^g(t \leftarrow s) - V^g(t \leftarrow r)V^g(r \leftarrow s) = g_t V(t \leftarrow s) g_s^{-1} - g_t V(t \leftarrow r) g_r^{-1} g_r V(r \leftarrow s) g_s^{-1} = g_t \Delta_V(t, s, r) g_s^{-1}$, and g_t, g_s are invertible. □

Lemma 3.2 suggests defining the removable deficit as the part that re-dressing can shrink, and the invariant as the orbital minimum of a flatness-distance. The following theorem shows the suggestion is empty.

Theorem 3.3 (Gauge trivialization)**.** *Let $V \in \mathcal{V}$ be arbitrary and let $w \in \mathbb{R}^n_{>0}$ be any strictly positive probability vector. For $\varepsilon \in (0, 1)$ define*

$$g_i^{(\varepsilon)} \;=\; \varepsilon^{\,i+1} I \;+\; \big(1 - \varepsilon^{\,i+1}\big)\, w\mathbf{1}^\top, \qquad i = 0, \ldots, m.$$

Then for all sufficiently small ε the re-dressing $g^{(\varepsilon)}$ is admissible for V, and as $\varepsilon \downarrow 0$,

$$V^{g^{(\varepsilon)}}(j \leftarrow i) \;\longrightarrow\; w\mathbf{1}^\top \qquad \text{for every pair } i < j.$$

The limit family, all of whose kernels equal the reset kernel $w\mathbf{1}^\top$, is flat. Consequently the closure of the orbit of every family contains flat families, and

$$\inf_{g \text{ admissible}} \operatorname{dist}\big(V^g, \mathcal{F}\mathrm{lat}\big) \;=\; 0 \qquad \text{for every } V \in \mathcal{V}.$$

Proof. Write $\varepsilon_i = \varepsilon^{\,i+1}$ and $g_i = \varepsilon_i I + (1 - \varepsilon_i) w\mathbf{1}^\top$. Since $\mathbf{1}^\top w = 1$ we have $\mathbf{1}^\top g_i = \mathbf{1}^\top$, and g_i is invertible with

$$g_i^{-1} \;=\; \tfrac{1}{\varepsilon_i}\Big(I - (1 - \varepsilon_i)\, w\mathbf{1}^\top\Big),$$

as direct multiplication verifies using $(w\mathbf{1}^\top)^2 = w\mathbf{1}^\top$. For $i < j$ put $\rho_{ij} = \varepsilon_j / \varepsilon_i = \varepsilon^{\,j-i} \leq \varepsilon$ and $A = V(j \leftarrow i)$. Using $\mathbf{1}^\top A = \mathbf{1}^\top$,

$$g_j A\, g_i^{-1} = \tfrac{1}{\varepsilon_i}\big(\varepsilon_j A + (1 - \varepsilon_j) w\mathbf{1}^\top\big)\big(I - (1 - \varepsilon_i) w\mathbf{1}^\top\big) = \rho_{ij}\, A \;-\; \rho_{ij}(1 - \varepsilon_i)\, Aw\mathbf{1}^\top \;+\; (1 - \varepsilon_j)\, w\mathbf{1}^\top,$$

where the cross terms were simplified with $\mathbf{1}^\top w = 1$ and $\mathbf{1}^\top A = \mathbf{1}^\top$. The (b, a) entry equals $(1 - \varepsilon_j) w_b + \rho_{ij}\big(A_{ba} - (1 - \varepsilon_i)(Aw)_b\big) \;\geq\; (1 - \varepsilon) \min_b w_b - \varepsilon$, which is positive for all pairs simultaneously once $\varepsilon < \min_b w_b / (1 + \min_b w_b)$; hence $g^{(\varepsilon)}$ is admissible. Letting $\varepsilon \downarrow 0$ gives $\rho_{ij} \to 0$ and $V^{g^{(\varepsilon)}}(j \leftarrow i) \to w\mathbf{1}^\top$ for every pair. The reset family is flat because $(w\mathbf{1}^\top)(w\mathbf{1}^\top) = w\mathbf{1}^\top$. Since $\operatorname{dist}(\cdot, \mathcal{F}\mathrm{lat})$ is continuous and vanishes at the limit, the infimum over the orbit is 0. □

Corollary 3.4. *Let $\Psi : \mathcal{V} \to [0, \infty)$ be continuous, constant on admissible orbits, and zero on $\mathcal{F}\mathrm{lat}$. Then $\Psi \equiv 0$.*

Proof. Ψ is constant on each orbit, hence on its closure by continuity; by Theorem 3.3 the closure contains a flat family, where Ψ vanishes. □

Remark 3.5 (Invariant-theoretic reading). The re-dressing group is the subgroup of $\mathrm{GL}_n(\mathbb{R})^{m+1}$ fixing the covector $\mathbf{1}^\top$ at each time, a non-reductive group, and $t \mapsto g^{(\varepsilon)}$ with $\varepsilon = e^{-t}$ is a one-parameter subgroup along which every family is driven to the reset locus. In the language of geometric invariant theory, every point of $\mathcal{V}$ is unstable for this action and the flat reset families lie in every orbit closure, so the categorical quotient collapses and the ring of continuous invariants vanishing on $\mathcal{F}\mathrm{lat}$ is trivial, which is the content of Corollary 3.4. Replacing the invariant by a bounded-conditioning surrogate does not help: the computations of Section 8 show that any such surrogate depends on the bound and is not constant on orbits. The failure is structural, and the remedy of the next section abandons the quotient for a metric projection onto a fixed model set, a construction to which orbit degeneration is irrelevant.

4 Dilation categories and the liftability profile

4.1 Lifts, compressions, and the model sets

Definition 4.1 (Dilation category and lifts)**.** Fix $d \geq n$. A *d-state flat lift* over the grid is a triple $\theta = (\pi, (M_k)_{k=1}^m, \varphi)$ where π is a probability vector on $[d]$, each M_k is a column-stochastic $d \times d$ matrix, the transport upstairs is $W_\theta(j \leftarrow i) = M_j M_{j-1} \cdots M_{i+1}$, and $\varphi : [d] \to [n]$ is a surjection, the *labeling*. The *time-homogeneous category* $\mathfrak{D}^{\mathrm{hom}}$ requires $M_1 = \cdots = M_m$; the *time-inhomogeneous category* $\mathfrak{D}^{\mathrm{inhom}}$ imposes no such constraint. Write $p_i(\theta) = W_\theta(i \leftarrow 0)\pi \in \mathbb{R}^d$ for the hidden law at time t_i and $\mu_i(\theta) \in \mathbb{R}^n$, $\mu_i(\theta)_a = \sum_{u:\varphi(u)=a} p_i(\theta)_u$, for the observed law. Let Θ_d^+ be the set of lifts (in the declared category) with $\mu_i(\theta)_a > 0$ for all i, a.

Definition 4.2 (Compression and model sets)**.** For $\theta \in \Theta_d^+$ the *compression* $\kappa(\theta) \in \mathcal{V}$ is the family of block-conditional kernels of the lifted chain:

$$\kappa(\theta)(j \leftarrow i)_{ba} \;=\; \mathbb{P}_\theta\big(\varphi(X_{t_j}) = b \,\big|\, \varphi(X_{t_i}) = a\big) \;=\; \frac{\mathbf{1}_b^\top W_\theta(j \leftarrow i)\,\mathrm{diag}(\mathbf{1}_a)\,p_i(\theta)}{\mu_i(\theta)_a}, \tag{6}$$

where $\mathbf{1}_a \in \{0,1\}^d$ indicates the block $\varphi^{-1}(a)$ and (X_{t_i}) is the hidden chain with initial law π. The *model set* in the declared category is

$$\mathcal{F}_d \;:=\; \mathrm{cl}\big(\kappa(\Theta_d^+)\big) \subseteq \mathcal{V},$$

and the *liftability profile* of $V \in \mathcal{V}$ is $h_d(V) = \mathrm{dist}(V, \mathcal{F}_d)$ as in (3), computed in the metric (4). When the category must be displayed we write $\mathcal{F}_d^{\mathrm{hom}}$, h_d^{hom}, and likewise for inhom.

The compression (6) is exactly the transition family observed when a Markov system is watched through the lumping φ; families of this form are the classical source of Chapman–Kolmogorov violations [10]. The closure in Definition 4.2 adjoins the boundary families arising when a block mass degenerates; Section 8 shows this boundary is genuinely touched by projections of non-realizable data.

Proposition 4.3 (Compactness and attainment)**.** *$\mathcal{F}_d$ is a nonempty compact subset of $\mathcal{V}$, and for every $V \in \mathcal{V}$ the distance $h_d(V)$ is attained at some $W^* \in \mathcal{F}_d$.*

Proof. $\mathcal{F}_d \subseteq \mathcal{V}$ is bounded, and it is closed by construction; a closed subset of the compact set $\mathcal{V}$ is compact. Nonemptiness: any flat family with a bijective-after-merging labeling lies in it, for instance the compression of the lift that places all hidden states of a block on identical dynamics. Distance to a nonempty compact set is attained. □

Proposition 4.4 (Nesting and monotonicity)**.** *In either category, $\mathcal{F}_d \subseteq \mathcal{F}_{d+1}$ for all $d \geq n$; consequently $h_{d+1}(V) \leq h_d(V)$ for every V.*

Proof. Let $\theta = (\pi, (M_k), \varphi) \in \Theta_d^+$ and fix a hidden state u. Define θ' on $[d+1]$ by splitting u into u', u'' with $\varphi'(u') = \varphi'(u'') = \varphi(u)$: the initial mass π_u is divided equally between u' and u''; each column of each M_k indexed by u' or u'' (as source) equals the old column of u with the row entry into u divided equally between the rows u' and u''; all other columns keep their entries, with the row into u likewise divided equally between u' and u''. Columns remain stochastic. The merge map $\psi : [d+1] \to [d]$ identifying u', u'' with u satisfies the strong lumpability criterion of Kemeny and Snell [10], because the two copies have identical columns as sources: the ψ-image of the primed chain is Markov with matrices (M_k) and initial law π. Hence the φ'-observed process of θ' coincides in law with the φ-observed process of θ, so $\kappa(\theta') = \kappa(\theta)$, and the block masses are unchanged and positive. Thus $\kappa(\Theta_d^+) \subseteq \kappa(\Theta_{d+1}^+)$, and closures preserve the inclusion. Homogeneity of the steps is preserved by the construction, so the argument applies verbatim in $\mathfrak{D}^{\mathrm{hom}}$. □

4.2 Semialgebraicity, stability, and weight-independence

Proposition 4.5 (Semialgebraicity and stratification)**.** *$\mathcal{F}_d$ is a compact semialgebraic subset of $\mathcal{V}$, and $V \mapsto h_d(V)^2$ is a continuous semialgebraic function. In particular $\mathcal{F}_d$ admits a Whitney stratification into finitely many smooth semialgebraic strata.*

Proof. Θ_d^+ is semialgebraic (polynomial equalities and strict inequalities), and each entry of $\kappa(\theta)$ in (6) is a rational function of θ whose denominator $\mu_i(\theta)_a$ is positive on Θ_d^+; over the finitely many labelings φ the graph of κ is semialgebraic, hence so is its image by the Tarski–Seidenberg theorem, and the closure of a semialgebraic set is semialgebraic [2, Ch. 2]. The squared distance $h_d^2(V) = \min_{W \in \mathcal{F}_d} \|V - W\|^2$ is then semialgebraic as a partial minimization of a polynomial over a semialgebraic set [2, Ch. 2], and it is 1-Lipschitz squared-root by Proposition 4.6, hence continuous. Existence of Whitney stratifications for semialgebraic sets is classical [2, Ch. 9]. □

Proposition 4.6 (Stability)**.** *For all $V, V' \in \mathcal{V}$ and all d, $\left|h_d(V) - h_d(V')\right| \leq \mathrm{dist}(V, V')$.*

Proof. Distance functions to a fixed set are 1-Lipschitz: for any $W \in \mathcal{F}_d$, $h_d(V) \leq \|V - W\| \leq \|V - V'\| + \|V' - W\|$; take the infimum over W and symmetrize. □

Remark 4.7. Proposition 4.6 is the estimation theorem in embryo: if $\widehat{V}_N$ is any estimator of the family with $\mathrm{dist}(\widehat{V}_N, V) \to 0$ (in probability, almost surely, or with a rate), then $h_d(\widehat{V}_N) \to h_d(V)$ in the same mode and with the same rate. The profile inherits, without further argument, whatever convergence theory the family estimator possesses.

Lemma 4.8 (Weight-independence of the strata)**.** *The sets $\{V : h_d(V) = 0\} = \mathcal{F}_d$, and hence the memory rank of Definition 5.1, do not depend on the choice of weights in* (4)*. Only the numerical values $h_d(V) > 0$ do, and any two admissible weight choices yield profiles equivalent up to two-sided multiplicative constants.*

Proof. All norms induced by (4) with strictly positive weights are equivalent on the finite-dimensional ambient space; equivalent norms have identical zero-distance sets and comparable distances. □

5 Memory rank, realization, and the realizability gap

Definition 5.1 (Memory rank)**.** $\mathrm{rk}_{\mathrm{mem}}(V) \;=\; \min\{\, d \geq n :\; h_d(V) = 0 \,\}$, with $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$ if no such d exists. By Proposition 4.4 the set of vanishing dimensions is upward closed, so the minimum is well defined, and by Lemma 4.8 it is independent of the metric weights. The rank is relative to the declared dilation category.

Example 5.2 (The dimension-matched charge is distance to flat). For $d = n$ every labeling is a bijection, compressions of bijectively labeled lifts are the lifted families themselves up to relabeling, and relabelings of flat families are flat; hence $\mathcal{F}_n^{\mathrm{inhom}} = \mathcal{F}\mathrm{lat}$ and

$$h_n(V) \;=\; \mathrm{dist}(V, \mathcal{F}\mathrm{lat}).$$

The bottom of the profile is the metric size of the Chapman–Kolmogorov failure itself, the quantity that plays the role of the family's charge in the companion balance-law analysis of coarse-graining.

Proposition 5.3 (Łojasiewicz comparison with the deficit)**.** *There exist constants $L, C > 0$ and an exponent $\theta \in (0, 1]$, depending only on n, the grid, and the weights, such that for all $V \in \mathcal{V}$*

$$\tfrac{1}{L}\,\|\Delta_V\| \;\leq\; \mathrm{dist}(V, \mathcal{F}\mathrm{lat}) \quad \textit{and} \quad \mathrm{dist}(V, \mathcal{F}\mathrm{lat}) \;\leq\; C\,\|\Delta_V\|^{\theta}.$$

Consequently h_n and the integrated deficit norms of [18, 20], which are weighted L^2 norms of Δ_V, are two-sidedly comparable up to a Łojasiewicz exponent: each vanishes exactly when the other does, and each controls the other on the compact $\mathcal{V}$.

Proof. The map $V \mapsto \Delta_V$ is polynomial, hence Lipschitz on the compact $\mathcal{V}$, and vanishes on $\mathcal{F}\mathrm{lat}$; the first inequality follows by evaluating at a nearest flat family. The second is the Łojasiewicz inequality applied to the semialgebraic function $f(V) = \|\Delta_V\|^2$, whose zero set is $\mathcal{F}\mathrm{lat}$: on a compact semialgebraic set, $\mathrm{dist}(V, f^{-1}(0)) \leq C|f(V)|^{\theta/2}$ for some $C > 0$ and $\theta \in (0, 1]$ [2, Ch. 2]. □

5.1 Necessity: the marginal-consistency defect

Lemma 5.4. *Every $W \in \mathcal{F}_d$ (either category) admits probability vectors $\mu_0, \ldots, \mu_m \in \Delta^{n-1}$ with*

$$W(j \leftarrow i)\,\mu_i \;=\; \mu_j \qquad \textit{for all } i < j. \tag{7}$$

Proof. For $W = \kappa(\theta)$ with $\theta \in \Theta_d^+$, take $\mu_i = \mu_i(\theta)$, the observed laws; (7) is the law of total probability. For $W \in \mathrm{cl}\,\kappa(\Theta_d^+)$, pick $\kappa(\theta_\ell) \to W$, pass to a subsequence along which the vectors $\mu_i(\theta_\ell)$ converge in the compact simplex, and take limits in (7). □

Definition 5.5. The *consistency defect* of $V \in \mathcal{V}$ is

$$\kappa_c(V) \;=\; \min_{\mu \in (\Delta^{n-1})^{m+1}} \; \max_{i<j} \; \big\| V(j \leftarrow i)\mu_i - \mu_j \big\|_1.$$

The minimum is attained by compactness, and $V \mapsto \kappa_c(V)$ is continuous.

Theorem 5.6 (Realizability gap)**.** *There is a constant $C_* > 0$, depending only on n, the grid, and the weights, such that for every $V \in \mathcal{V}$ and every $d \geq n$,*

$$h_d(V) \;\geq\; \frac{\kappa_c(V)}{C_*}.$$

In particular, if $\kappa_c(V) > 0$ then $h_d(V)$ is bounded away from zero uniformly in d and $\mathrm{rk}_{\mathrm{mem}}(V) = \infty$: no dilation of any dimension explains the family.

Proof. κ_c is Lipschitz in V: for fixed μ, $\|(V-W)(j \leftarrow i)\mu_i\|_1 \le \|V-W\|_{1\to 1,ij} \le C_*\,\mathrm{dist}(V,W)$ with C_* a norm-equivalence constant, for instance $C_* = n\left(\min_{i<j,a} p_{ij}\pi_i(a)\right)^{-1/2}$; taking maxima and minimizing over μ gives $|\kappa_c(V)-\kappa_c(W)| \le C_*\,\mathrm{dist}(V,W)$. By Lemma 5.4, $\kappa_c \equiv 0$ on $\mathcal{F}_d$; evaluating at a nearest point W^* gives $\kappa_c(V) \le C_*\,h_d(V)$. □

5.2 Sufficiency: the realization theorem

Call $V \in \mathcal{V}$ *process-induced* if there exists a stochastic process $(X_{t_i})_{i=0}^m$ on $[n]$ with $\mathbb{P}(X_{t_i}=a)>0$ for all i,a and $\mathbb{P}(X_{t_j}=b \mid X_{t_i}=a) = V(j \leftarrow i)_{ba}$ for all $i<j$ and all a,b.

Theorem 5.7 (Realization and termination of the profile)**.** *Work in $\mathfrak{D}^{\mathrm{inhom}}$. If V is process-induced, then $V \in \kappa(\Theta_D^+)$ exactly, hence $h_D(V)=0$ and*

$$\mathrm{rk}_{\mathrm{mem}}(V) \;\le\; D \;:=\; \sum_{k=0}^{m} N_k \;\le\; \sum_{k=0}^{m} n^{k+1},$$

where N_k is the number of length-$(k{+}1)$ observed histories of positive probability. Conversely, every element of every $\mathcal{F}_d$ is a limit of process-induced families. Consequently

$$\big\{\, V : \lim_{d\to\infty} h_d(V) = 0 \,\big\} \;=\; \mathrm{cl}\{\,\textit{process-induced families}\,\},$$

and on this set the limit is attained at the finite dimension D.

Proof. Sufficiency. Let $H_k \subseteq [n]^{k+1}$ be the set of positive-probability histories $(x_0,\dots,x_k)$ of the process, $N_k = |H_k|$, and take the hidden space $[D] = \bigsqcup_{k=0}^m H_k$, a disjoint union of layers. The hidden chain occupies layer k at time t_k: the initial law places mass $\mathbb{P}(X_{t_0}=x_0)$ on each $(x_0)\in H_0$; the step matrix M_{k+1} sends $h=(x_0,\dots,x_k)\in H_k$ to $h\cdot b = (x_0,\dots,x_k,b)\in H_{k+1}$ with probability $\mathbb{P}(X_{t_{k+1}}=b \mid X_{t_0}{=}x_0,\dots,X_{t_k}{=}x_k)$, a well-defined conditional since h has positive probability, and sends every state outside layer k to an arbitrary fixed state of layer $k{+}1$ (these sources carry zero mass at time t_k, so the choice is immaterial); columns are stochastic by construction. The labeling φ sends a history to its last symbol on each layer. The history process $(X_{t_0},\dots,X_{t_k})_k$ is Markov with these steps, its law restricted to the layers reproduces the path law of (X_{t_i}), the observed block masses at time t_k are $\mathbb{P}(X_{t_k}=a)>0$, and the block conditionals (6) are exactly $\mathbb{P}(X_{t_j}=b \mid X_{t_i}=a) = V(j\leftarrow i)_{ba}$. Hence $V=\kappa(\theta)$ with $\theta\in\Theta_D^+$.

Necessity. For $\theta\in\Theta_d^+$ the compressed family $\kappa(\theta)$ is process-induced by the observed process $\varphi(X_{t_i})$, whose one-time marginals are the positive block masses; elements of the closure are limits of such families. The final display combines the two directions with Proposition 4.4, and attainment at D is the first part. □

Remark 5.8. Theorem 5.7 is a statement about $\mathfrak{D}^{\mathrm{inhom}}$; the history lift is intrinsically time-layered. In $\mathfrak{D}^{\mathrm{hom}}$ the profile of a process-induced family need not terminate at the same dimension, and Section 7 quantifies the gap. The bound D is crude and the numerical evidence of Section 8 shows the true rank is often far smaller; sharpening D toward the positive-realization bounds of [1, 21] is Open Problem (iii).

6 First-order theory at the projection

Fix d and $V \notin \mathcal{F}_d$, and let $W^* \in \mathcal{F}_d$ attain $h_d(V)$ (Proposition 4.3). Because $\mathcal{F}_d$ is closed, the standard necessary condition of variational analysis applies: the residual $V-W^*$ belongs to the

regular normal cone $\widehat{N}_{\mathcal{F}_d}(W^*)$, equivalently

$$\langle V - W^*,\, Z - W^* \rangle \;\le\; o(\|Z - W^*\|) \qquad \text{as } Z \to W^*,\; Z \in \mathcal{F}_d, \tag{8}$$

see [15, Ch. 6]. Refinements follow from the stratification of Proposition 4.5. Let $S \subseteq \mathcal{F}_d$ be the smooth stratum containing W^*. Since $S \subseteq \mathcal{F}_d$, normal cones reverse the inclusion, $\widehat{N}_{\mathcal{F}_d}(W^*) \subseteq N_S(W^*)$, so (8) implies the classical orthogonality

$$V - W^* \;\perp\; T_S(W^*), \tag{9}$$

and the residual decomposes trivially along $T_S(W^*) \oplus N_S(W^*)$ with vanishing tangential part: at the projection, all of the deficit that the stratum's directions could absorb has been absorbed, and what remains is normal. When W^* lies on a boundary stratum, (8) is the correct statement and the tangent object is the tangent cone; the Moreau decomposition of $V - W^*$ against the tangent and polar cones of $\mathcal{F}_d$ at W^* then replaces (9). Section 8 verifies (9) numerically at interior projections to the reported stationarity tolerances and records boundary contact for non-realizable data.

Two standard facts complete the local picture. First, the set of V with non-unique projection onto the closed set $\mathcal{F}_d$ has Lebesgue measure zero, since the squared distance is locally semiconcave and hence differentiable almost everywhere, and differentiability forces uniqueness [15, Ch. 6]; generic data therefore have a unique nearest explanation of each dimension. Second, the number of smooth critical points of the squared-distance function on the Zariski closure of a stratum is bounded by its Euclidean distance degree [6]; the multistart landscapes observed in Section 8 are samples of exactly this critical-point structure, and computing the ED degrees of the strata of $\mathcal{F}_d$ is Open Problem (iv).

7 Category dependence

Proposition 7.1. *For every $d \ge n$, $\mathcal{F}_d^{\mathrm{hom}} \subseteq \mathcal{F}_d^{\mathrm{inhom}}$ and hence $h_d^{\mathrm{inhom}} \le h_d^{\mathrm{hom}}$ pointwise. The inequality can be strict: there exist families V with $h_d^{\mathrm{inhom}}(V) = 0 < h_d^{\mathrm{hom}}(V)$, and the gap admits an explicit lower bound.*

Proof. The inclusion of categories gives the inclusion of model sets and the inequality. For strictness take $d = n$, so labelings are bijections and, as in Example 5.2, $\mathcal{F}_n^{\mathrm{inhom}} = \mathcal{F}\mathrm{lat}$ while every $W \in \mathcal{F}_n^{\mathrm{hom}}$ satisfies $W(1 \leftarrow 0) = W(2 \leftarrow 1)$, a closed condition preserved under the closure. Choose V flat with one-step kernels $A = V(1 \leftarrow 0) \ne B = V(2 \leftarrow 1)$ and $V(j \leftarrow i)$ the corresponding products. Then $h_n^{\mathrm{inhom}}(V) = 0$, while for any $W \in \mathcal{F}_n^{\mathrm{hom}}$ with common one-step kernel M, using only the two weighted summands for the pairs $(0,1)$ and $(1,2)$ in (4),

$$\mathrm{dist}(V,W)^2 \;\ge\; \min(p_{01}, p_{12})\big(\|A - M\|_\pi^2 + \|B - M\|_\pi^2\big) \;\ge\; \tfrac{1}{2}\min(p_{01}, p_{12})\,\|A - B\|_\pi^2,$$

by the parallelogram identity in the inner-product norm, the minimum over M being attained at the stochastic matrix $(A+B)/2$. Hence $h_n^{\mathrm{hom}}(V) \ge \big(\tfrac{1}{2}\min(p_{01}, p_{12})\big)^{1/2}\|A - B\|_\pi > 0$. □

The phenomenon is the temporal analog of the dependence of lift size on the lifting cone in extension-complexity theory [22]: the same observed object has different minimal explanations, and different distances to explanation, in different categories, and time-inhomogeneity can substitute for hidden dimension. Section 8 measures the exchange rate on concrete families.

8 Computations

All computations use $n = 2$ observed states, the grid $\mathbb{T} = \{0, 1, 2, 3, 4\}$, and uniform weights $p_{ij} = 1$, $\pi_i = \mathbf{1}$ in (4); by Lemma 4.8 the zero strata and ranks reported are weight-independent. Distances h_d are computed by minimizing $\mathrm{dist}(V, \kappa(\theta))^2$ over softmax charts of Θ_d^+ with multistart quasi-Newton descent (L-BFGS-B, up to fourteen starts per cell, Powell polish on the incumbent), one representative labeling per block-size class. Reported values are therefore certified upper bounds on h_d, with multistart dispersion and first-order stationarity (at or below 10^{-7} in every cell, typically 10^{-9}) as evidence of tightness; entries displayed as 0.000000 are below 5×10^{-7}. The coherence identity (5) and the covariance of Lemma 3.2 hold on every family tested to 2×10^{-16}. Code reproducing every number is available from the author.

8.1 Recovery of hidden dimension

Three families: a flat 2-state chain observed in full; the compression of a randomly generated 4-state homogeneous chain through a two-and-two lumping; and the compression of a 3-state chain through a two-and-one lumping. Profiles in $\mathfrak{D}^{\mathrm{hom}}$:

Table 1: Liftability profiles, homogeneous category. Zeros are exact to displayed precision.

Family	h_2	h_3	h_4	h_5
Flat 2-state	0.000000			
Lumped $4 \to 2$	0.116164	0.026680	0.000000	≤ 0.000222
Lumped $3 \to 2$	0.101877	0.000000	0.000000	

The profile does not merely decrease; it vanishes exactly at the generating dimension, so $\mathrm{rk}_{\mathrm{mem}} = 4$ and $\mathrm{rk}_{\mathrm{mem}} = 3$ are recovered, and below the rank the charge is decisively positive. All projections in Table 1 are attained at interior points (minimal block masses between 0.12 and 0.40), and (9) holds to the stated stationarity.

8.2 The realizability plateau

A family assembled from ten independent uniformly random stochastic matrices, one per pair, is the conditional family of no process; Theorem 5.6 predicts a profile bounded away from zero. Measured, in $\mathfrak{D}^{\mathrm{hom}}$: $h_2 = 1.319266$, $h_3 = 1.174508$, $h_4 = 0.963165$, $h_5 = 0.910255$. The plateau is unambiguous against the exact zeros of Table 1, and the projections for this family repeatedly contact the boundary strata of $\mathcal{F}_d$ (block masses down to 10^{-13}), the regime where (8) rather than (9) governs.

8.3 Category exchange rate

On the lumped $4 \to 2$ family, passing from $\mathfrak{D}^{\mathrm{hom}}$ to $\mathfrak{D}^{\mathrm{inhom}}$ moves h_2 from 0.116164 to 0.091166 and h_3 from 0.026680 to 0.000752: time-inhomogeneity absorbs almost all of what the fourth hidden state was needed for. On the random family, h_3 moves from 1.174508 to 0.841519: the realizability floor itself is category-relative. The declared category is load-bearing, exactly as Proposition 7.1 asserts.

8.4 The gauge route, numerically

Two illustrations of Section 3 on generic families with $\|\Delta_V\|^2$ of order one. Along the explicit path of Theorem 3.3, the squared deficit of a family with initial value 0.677 descends below 10^{-11} while every transformed kernel remains entrywise nonnegative, with the transformed family visibly approaching the reset form. Guarding the re-dressings with a hard determinant floor of 0.25 does not restore an invariant: the constrained minimizer routes through condition numbers up to 10^8, removes ninety-five percent of the squared deficit, and returns different values, 0.2063 against 0.2168 on the same family after an admissible re-dressing, confirming Corollary 3.4 in its practical form: every bounded surrogate of the orbital minimum depends on the bound and fails orbit-invariance.

9 Open problems

(i) *Selection bridge.* The selection functionals of [20] extract predictive values from a family without a generator by reweighting proposed paths against a surrogate. Conjecture: for process-induced V, the selection-functional readout of V agrees with the readout of the compression of its projection onto $\mathcal{F}_d$ up to an error controlled by $h_d(V)$, with equality on the zero stratum. Either the identity or an h_d-controlled bound would connect the estimation theory of [20] to the geometry developed here.

(ii) *Natural contractions.* Classify the contractions of the deficit Δ_V that are natural with respect to the morphisms of the dilation category, lumpings and dilations, and compatible with the coherence identity (5). Example 5.2 exhibits one, the dimension-matched charge h_n; a classification would determine whether it is essentially unique.

(iii) *Sharp rank bounds.* Replace the history-lift bound D of Theorem 5.7 by bounds in the spirit of positive realization theory [1, 21], and characterize the families on the boundary strata of $\mathcal{F}_d$.

(iv) *ED degrees.* Compute or bound the Euclidean distance degrees [6] of the strata of $\mathcal{F}_d$ as functions of (n, d, m) and the category, thereby bounding the multistart complexity of the profile.

(v) *Behavior under lumpings.* For a lumping Λ compatible with the law of the process, decompose $h_n(\Lambda V)$ into a contractive image of the fine charge and a production term with an identified zero set. The companion analysis of this question is the subject of separate work; the geometric ingredients, the metric (4) and the projection of Section 6, are those developed here.

(vi) *Continuous grids.* Formulate the profile for families over an interval, where $\mathcal{V}$ is infinite-dimensional, compactness of $\mathcal{F}_d$ must be argued in an appropriate topology, and the grid-dependence of $\mathrm{rk}_{\mathrm{mem}}$ becomes a refinement question.

Declarations

AI assistance was used in the preparation of this manuscript. All theoretical frameworks, methodological approaches, and conclusions are the author's original work; the author reviewed all content and assumes responsibility for it.

Competing interests

The author declares no competing interests. Funding information appears in the footnote to the title.

Data and code availability

All numerical results of Section 8 are reproduced by code available from the author on request; every family, metric, and projection is specified in the text. The families are generated as stated, and all reported quantities follow from the definitions given.

References

[1] B. D. O. Anderson, The realization problem for hidden Markov models. *Mathematics of Control, Signals and Systems* **12** (1999), 80–120.

[2] J. Bochnak, M. Coste, and M.-F. Roy, *Real Algebraic Geometry*. Ergebnisse der Mathematik 36, Springer, 1998.

[3] P. Breiding, K. Kohn, and B. Sturmfels, *Metric Algebraic Geometry*. Oberwolfach Seminars 53, Birkhäuser, 2024.

[4] H.-P. Breuer, E.-M. Laine, and J. Piilo, Measure for the degree of non-Markovian behavior of quantum processes in open systems. *Physical Review Letters* **103** (2009), 210401.

[5] L. Broux and L. Zambotti, The sewing lemma for $0 < \gamma \leq 1$. *Journal of Functional Analysis* **283**(10) (2022), 109644.

[6] J. Draisma, E. Horobeţ, G. Ottaviani, B. Sturmfels, and R. R. Thomas, The Euclidean distance degree of an algebraic variety. *Foundations of Computational Mathematics* **16** (2016), 99–149.

[7] D. Feyel, A. de La Pradelle, and G. Mokobodzki, A non-commutative sewing lemma. *Electronic Communications in Probability* **13** (2008), 24–34.

[8] M. Gubinelli, Controlling rough paths. *Journal of Functional Analysis* **216** (2004), 86–140.

[9] B. L. Ho and R. E. Kalman, Effective construction of linear state-variable models from input/output functions. *Regelungstechnik* **14** (1966), 545–548.

[10] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*. Van Nostrand, 1960.

[11] K. Lê, A stochastic sewing lemma and applications. *Electronic Journal of Probability* **25** (2020), paper no. 38.

[12] N. Li and X. Gao, A low-regularity semigroup sewing lemma via quotient structures. arXiv:2606.16164 (2026).

[13] S. Milz and K. Modi, Quantum stochastic processes and quantum non-Markovian phenomena. *PRX Quantum* **2** (2021), 030201.

[14] Á. Rivas, S. F. Huelga, and M. B. Plenio, Entanglement and non-Markovianity of quantum evolutions. *Physical Review Letters* **105** (2010), 050403.

[15] R. T. Rockafellar and R. J.-B. Wets, *Variational Analysis*. Grundlehren der mathematischen Wissenschaften 317, Springer, 1998.

[16] J. D. Stasheff, Homotopy associativity of H-spaces I. *Transactions of the American Mathematical Society* **108** (1963), 275–292.

[17] B. D. Summers, The incompleteness of the instantaneous state and the status of Chapman–Kolmogorov divisibility: analysis from limits, relativity, and renormalization. SSRN, `https://doi.org/10.2139/ssrn.6922380` (2026).

[18] B. D. Summers, Koopman spectral theory for indivisible stochastic processes. SSRN, `https://doi.org/10.2139/ssrn.6733181` (2026).

[19] B. D. Summers, Convex perturbation and the suppression of the indivisibility deficit. SSRN, `https://doi.org/10.2139/ssrn.6812461` (2026).

[20] B. D. Summers, Selection functionals on path space for stochastic processes without a generator. SSRN, https://doi.org/10.2139/ssrn.6922758 (2026).

[21] M. Vidyasagar, The complete realization problem for hidden Markov models: a survey and some new results. *Mathematics of Control, Signals and Systems* **23** (2011), 1–65.

[22] M. Yannakakis, Expressing combinatorial optimization problems by linear programs. *Journal of Computer and System Sciences* **43** (1991), 441–466.